

09: Intro to Multiple Linear Regression

Stat 230 – Spring 2026

Note

This is the tutorial version of the R activity. If you prefer to use .qmd and RStudio directly, you can use <https://aluby.github.io/stat230/activities/09-mlr>

1 Data

- In an effort to understand the important aspects of a satisfactory supervisor, clerical employees at a large financial organization were asked to rate their immediate supervisor.
- The survey questions were designed to measure overall performance of the supervisor, as well as six additional characteristics.
- Employees were asked to rate the following statements on a scale from 0 to 100 (0 = “completely disagree”, 100 = “completely agree”)

Note: this data comes from the textbook

Variable	Description
rating	Overall rating of supervisor performance
complaints	Score for “Your supervisor handles employee complaints appropriately.”
privileges	Score for “Your supervisor allows special privileges.”
learn	Score for “Your supervisor provides opportunities to learn new things.”
raises	Score for “Your supervisor bases raises on performance.”
critical	Score for “Your supervisor is too critical of poor performance.”
advance	Score for “I am not satisfied with the rate I am advancing in the company?”

Before looking at the data, how do you think each predictor variable will be related to rating? What do you think will have the strongest relationship? The weakest?

1.1 Loading data

To load the supervisor data set, run the following code chunk:

```
library(dplyr)
library(GGally)

supervisor <- read.csv("https://aluby.github.io/stat230/data/supervisor.csv")
```

1.2 EDA: Making a pairs plot

We will use the `ggpairs()` function in the `{GGally}` package to make the pairs plot. By default, it will include every variable in the dataset. The code below firsts `select()`s the columns we want to include (all of them). I have also moved the response variable to be the last in the list, so that it shows up on the Y axis.

```
supervisor |>
  select(complaints, privileges, learn, raises, critical, advance, overall) |>
  ggpairs()
```

- Which predictor variables appear to be most closely related to `overall`?
- Which predictor variables appear to be most closely related to *each other*?

1.3 SLR

In the code chunk below, choose what you think will be the best *single* predictor and fit an SLR model. Interpret the intercept and slope coefficient and report the R^2 value.

```
supervisor_slr <- lm(overall ~ _____, data = supervisor)
summary(supervisor_slr)
```

1.4 MLR

In the code chunk below, use all predictors and fit an MLR model. Interpret the intercept and two of the slope coefficients and report the R^2 value.

```
supervisor_mlr <- lm(overall ~ _____, data = supervisor)
summary(supervisor_mlr)
```

2 Predicting responses

We can use most of our SLR tools for MLR. For example, here's some code I could use to calculate predictions. (Uses the bikes data)

```
predict(bikes_lm, newdata = data.frame(temp_actual = 90, windspeed = 6))
```

- Use this as starter code to predict the overall manager score for a manager with scores of 50% for all predictors
- What if the manager from the previous question decides to try to improve her critical score by 10 points (so 40% instead of 50%) and her complaints score by 10 points (so 60% instead of 50%). How does that change the prediction?

```
# your code here
```

2.1 Function quick reference

Function	Purpose
lm(formula, data)	Fit a MLR model
predict(model, newdata, interval, level, se.fit)	Make predictions and calculate intervals
data > select(vars) > ggpairs()	Make pairs plot for selected variables